

Table of Fourier Transforms

$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{ix\omega} d\omega$		$\hat{f}(\omega) = \mathcal{F}(f)(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$
1.	$\begin{cases} 1 & \text{if } x < a \\ 0 & \text{if } x > a \end{cases}$	$\sqrt{\frac{2}{\pi}} \frac{\sin a\omega}{\omega}$
2.	$\begin{cases} 1 & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$	$\frac{i (e^{-ib\omega} - e^{-ia\omega})}{\sqrt{2\pi}\omega}$
3.	$\begin{cases} 1 - \frac{ x }{a} & \text{if } x < a \\ 0 & \text{if } x > a \end{cases} \quad a > 0$	$2\sqrt{\frac{2}{\pi}} \frac{\sin^2(\frac{a\omega}{2})}{a\omega^2}$
4.	$\begin{cases} x & \text{if } x < a \\ 0 & \text{if } x > a \end{cases} \quad a > 0$	$i\sqrt{\frac{2}{\pi}} \frac{a\omega \cos(a\omega) - \sin(a\omega)}{\omega^2}$
5.	$\begin{cases} \sin x & \text{if } x < \pi \\ 0 & \text{if } x > \pi \end{cases}$	$i\sqrt{\frac{2}{\pi}} \frac{\sin(\pi\omega)}{\omega^2 - 1}$
6.	$\begin{cases} \sin(ax) & \text{if } x < b \\ 0 & \text{if } x > b \end{cases} \quad a, b > 0$	$i\sqrt{\frac{2}{\pi}} \frac{\omega \cos(b\omega) \sin(ab) - a \cos(ab) \sin(b\omega)}{\omega^2 - a^2}$
7.	$\frac{1}{a^2 + x^2}, a > 0$	$\sqrt{\frac{\pi}{2}} \frac{e^{-a \omega }}{a}$
8.	$\frac{x}{a^2 + x^2}, a > 0$	$-i\sqrt{\frac{\pi}{2}} \operatorname{sgn} \omega e^{-a \omega }$
9.	$\sqrt{\frac{2}{\pi}} \frac{a}{1 + a^2 x^2}, a > 0$	$e^{-\frac{ \omega }{a}}$
10.	$\frac{\sin ax}{x}, a > 0$	$\begin{cases} \sqrt{\frac{\pi}{2}} & \text{if } \omega < a \\ \frac{1}{2} \sqrt{\frac{\pi}{2}} & \text{if } \omega = a \\ 0 & \text{if } \omega > a \end{cases}$
11.	$\frac{4}{\sqrt{2\pi}} \frac{\sin^2(\frac{1}{2}ax)}{ax^2}, a > 0$	$\begin{cases} 1 - \frac{ \omega }{a} & \text{if } \omega < a \\ 0 & \text{if } \omega > a \end{cases}$
12.	$\frac{4}{\sqrt{2\pi}} \frac{\sin^2(ax) - \sin^2(\frac{1}{2}ax)}{ax^2}, a > 0$	$\begin{cases} 1 & \text{if } x < a \\ (-x + 2a)/a & \text{if } a < x < 2a \\ (x + 2a)/a & \text{if } a < x < 2a \\ 0 & \text{if } x > 2a \end{cases}$
13.	$e^{-a x }, a > 0$	$\sqrt{\frac{2}{\pi}} \frac{a}{a^2 + \omega^2}$
14.	$\begin{cases} e^{-ax} & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}, a > 0$	$\frac{1}{\sqrt{2\pi}} \frac{1}{a + i\omega}$
15.	$\begin{cases} 0 & \text{if } x > 0 \\ e^{ax} & \text{if } x < 0 \end{cases}, a > 0$	$\frac{1}{\sqrt{2\pi}} \frac{1}{a - i\omega}$
16.	$ x ^n e^{-a x }, a > 0, n > 0$	$\frac{\Gamma(n+1)}{\sqrt{2\pi}} \left(\frac{1}{(a - i\omega)^{1+n}} + \frac{1}{(a + i\omega)^{1+n}} \right)$

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17.	$e^{-\frac{a}{2}x^2}, a > 0$	$\frac{1}{\sqrt{a}} e^{-\frac{\omega^2}{2a}}$	
18.	$e^{-ax^2}, a > 0$	$\frac{1}{\sqrt{2a}} e^{-\frac{\omega^2}{4a}}$	
19.	$xe^{-\frac{a}{2}x^2}, a > 0$	$\frac{-i\omega}{a^{3/2}} e^{-\frac{\omega^2}{2a}}$	
20.	$x^2 e^{-\frac{a}{2}x^2}, a > 0$	$\frac{a - \omega^2}{a^{5/2}} e^{-\frac{\omega^2}{2a}}$	
21.	$x^3 e^{-\frac{a}{2}x^2}, a > 0$	$\frac{-i\omega(3a - \omega^2)}{a^{7/2}} e^{-\frac{\omega^2}{2a}}$	
22.	$e^{-\frac{x^2}{2}} H_n(x),$ H_n, nth Hermite polynomial	$(-1)^n i^n e^{-\frac{\omega^2}{2}} H_n(\omega)$	
23.	$J_0(x),$ Bessel function of order 0	$\begin{cases} \sqrt{\frac{2}{\pi}} \frac{1}{\sqrt{1-\omega^2}} & \text{if } \omega < 1 \\ 0 & \text{if } \omega > 1 \end{cases}$	
24.	$J_n(x),$ Bessel function of order $n \geq 0$	$\begin{cases} \sqrt{\frac{2}{\pi}} \frac{(-i)^n}{\sqrt{1-\omega^2}} T_n(\omega) & \text{if } \omega < 1 \\ 0 & \text{if } \omega > 1 \end{cases}$	
		$T_n, \text{Chebyshev polynomial of degree } n.$	
Special Transforms			
25.	$\mathcal{F}(\delta_0(x))(\omega) = \frac{1}{\sqrt{2\pi}}$	27.	$\mathcal{F}\left(\sqrt{\frac{2}{\pi}} \frac{1}{x}\right)(\omega) = -i \operatorname{sgn} \omega$
26.	$\mathcal{F}(\delta_0(x-a))(\omega) = \frac{1}{\sqrt{2\pi}} e^{-ia\omega}$	28.	$\mathcal{F}(e^{iax})(\omega) = \sqrt{2\pi} \delta_0(\omega-a)$
Operational Properties			
29.	$\mathcal{F}(af+bg)(\omega) = a\mathcal{F}(f) + b\mathcal{F}(g)$	36.	$\mathcal{F}(fg)(\omega) = \mathcal{F}(f) * \mathcal{F}(g)(\omega)$
30.	$\mathcal{F}(f')(\omega) = i\omega \mathcal{F}(f)(\omega)$	37.	$\mathcal{F}(f(x-a))(\omega) = e^{-ia\omega} \mathcal{F}(f)(\omega)$
31.	$\mathcal{F}(f'')(\omega) = -\omega^2 \mathcal{F}(f)(\omega)$	38.	$\mathcal{F}(e^{iax} f(x))(\omega) = \mathcal{F}(f)(\omega-a)$
32.	$\mathcal{F}(f^{(n)})(\omega) = (i\omega)^n \mathcal{F}(f)(\omega)$	39.	$\mathcal{F}(\cos(ax)f(x))(\omega) = \frac{\mathcal{F}(f)(\omega-a) + \mathcal{F}(f)(\omega+a)}{2}$
33.	$\mathcal{F}(xf(x))(\omega) = i \frac{d}{d\omega} \mathcal{F}(f)(\omega)$	40.	$\mathcal{F}(\sin(ax)f(x))(\omega) = \frac{\mathcal{F}(f)(\omega-a) - \mathcal{F}(f)(\omega+a)}{2i}$
34.	$\mathcal{F}(x^n f(x))(\omega) = i^n \frac{d^n}{d\omega^n} \mathcal{F}(f)(\omega)$	41.	$\mathcal{F}(f(ax))(\omega) = \frac{1}{ a } \mathcal{F}(f)\left(\frac{\omega}{a}\right), a \neq 0$
35.	$\mathcal{F}(f * g)(\omega) = \mathcal{F}(f)(\omega) \mathcal{F}(g)(\omega)$	42.	$f(x) = \mathcal{F}(\widehat{f})(-x), \mathcal{F}(\mathcal{F}(f)) = f(-x)$

Source:

Nakhlé H. Asmar, Partial Differential Equations with Fourier Series and Boundary Value Problems 2nd Edition, 2004